

# Machine learning-driven integrated safety and security risk assessment for industrial control systems in manufacturing to automate compliance reporting for emerging regulations.

Manufacturing × Regulatory compliance and reporting × Machine learning · OS105 v1 · refreshed 2026-08-18

| ATTRACTIVENESS                   | RIGHT TO WIN                         | COMPETITION                              | SERVICEABLE MARKET                           | HORIZON                                             | PORTFOLIO               |
|----------------------------------|--------------------------------------|------------------------------------------|----------------------------------------------|-----------------------------------------------------|-------------------------|
| <b>38</b><br>is the world moving | <b>74</b><br>can we play, can we win | <b>MEDIUM</b><br>0 named in the evidence | <b>€309m</b><br>observed evidence · per year | <b>LATER</b><br>research and standards<br>activi... | <b>L1</b><br>14 signals |

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

## The opportunity

This opportunity space addresses the need for integrated safety and security risk assessment in industrial control systems, driven by emerging regulations and the recognition that purely safety or security assessments are insufficient. For manufacturing operators, Orange can deliver a machine learning-driven assessment that automates compliance reporting, leveraging our AI, cybersecurity, and IoT expertise. The time is now because cross-domain risk propagation is a recognized gap, and regulatory pressure is mounting.

## Market opportunity

| Method                                             | Total (TAM)                      | Serviceable (SAM)                | Obtainable (SOM)                | Confidence |
|----------------------------------------------------|----------------------------------|----------------------------------|---------------------------------|------------|
| Bottom-up: enterprises x adoption x contract value | <b>€431m</b><br>€77.6m – €2.3bn  | <b>€309m</b><br>€55.7m – €1.7bn  | <b>€14.9m</b><br>€2.7m – €79.8m | observed   |
| Observed: tendered contract value, annualised      | <b>€20.9m</b><br>€20.9m – €56.6m | <b>€20.9m</b><br>€20.9m – €56.6m | <b>€1.0m</b><br>€1.0m – €2.7m   | observed   |

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

### Bottom-up: enterprises x adoption x contract value

| Factor                                             | Value                                                              | Basis      | Source                                        | As of                  |
|----------------------------------------------------|--------------------------------------------------------------------|------------|-----------------------------------------------|------------------------|
| Enterprises in Manufacturing (EU27_2020)           | 238,122 enterprises                                                | observed   | Eurostat · sbs_sc_ovw                         | 2024                   |
| Enterprises adopting Machine learning              | 3.3 % of enterprises (10+ employees)                               | observed   | Eurostat · isoc_eb_ain2 · E_AI_TML            | 2025                   |
| Annual value of one engagement                     | €337k per enterprise per year                                      | observed   | TED (Tenders Electronic Daily) · ted          | 2026-05-19..2026-08-05 |
| Size-weighted buyer base                           | 39,220 enterprise-equivalents at large-enterprise engagement value | assumption | config/sizing.yaml · size_class_value_weights | size-2026-08-a         |
| Assumed obtainable share of the serviceable market | 4.8 % of SAM                                                       | assumption | config/sizing.yaml · obtainable_share         | size-2026-08-a         |

**Read this before quoting the number.** Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

### Observed: tendered contract value, annualised

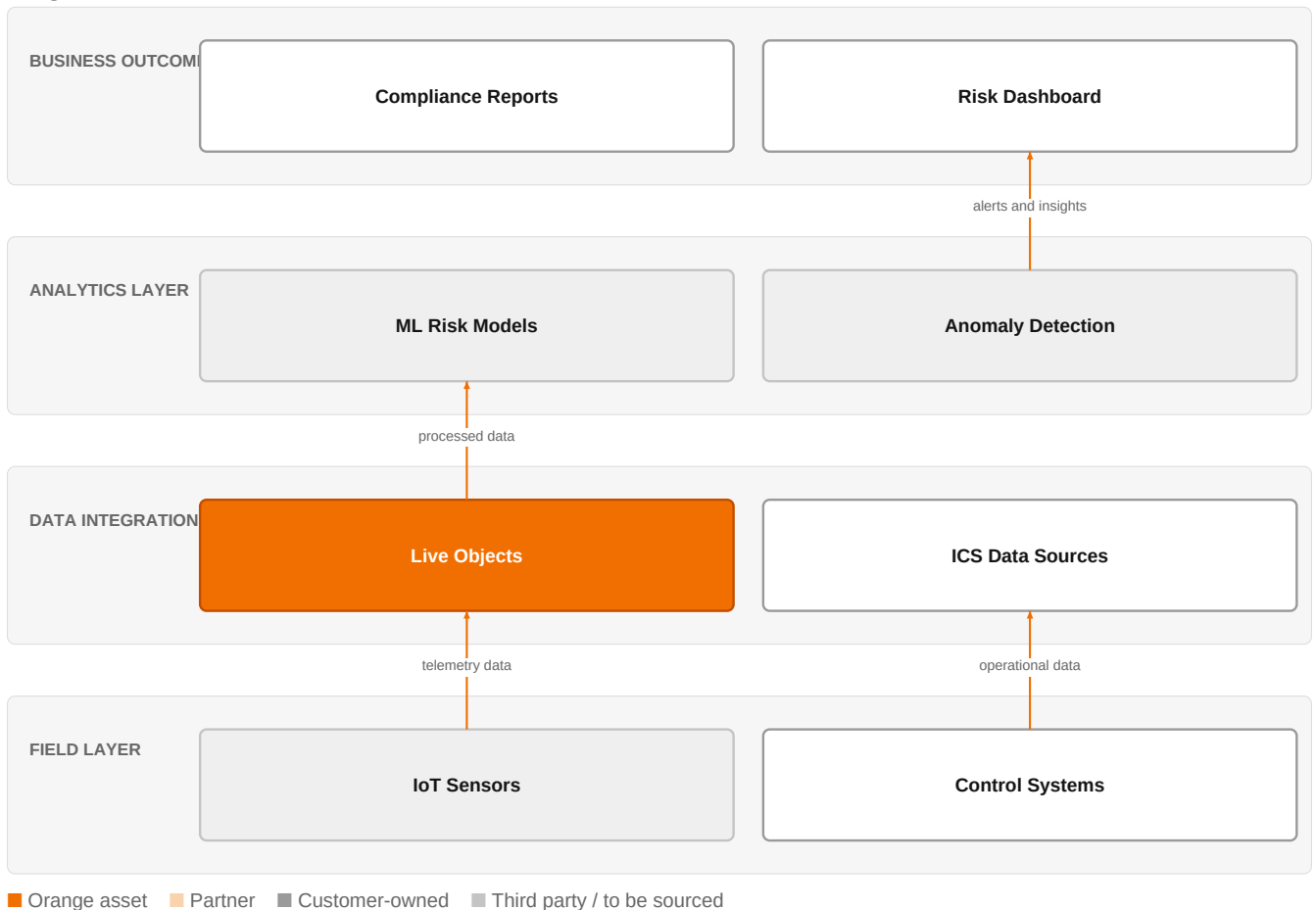
| Factor                                             | Value           | Basis      | Source                                | As of                  |
|----------------------------------------------------|-----------------|------------|---------------------------------------|------------------------|
| Tendered contract value in matching CPV categories | €20.9m per year | observed   | TED (Tenders Electronic Daily) · ted  | 2026-05-19..2026-08-05 |
| Assumed obtainable share of the serviceable market | 4.8 % of SAM    | assumption | config/sizing.yaml · obtainable_share | size-2026-08-a         |

**Read this before quoting the number.** Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size.

## The solution, in one picture

### Integrated Risk Assessment for ICS



This diagram shows how Orange integrates IoT data, machine learning, and customer systems to deliver automated compliance reporting and risk visibility.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

## What has changed

Industrial control systems face dual risks in cyber and physical domains, with risks propagating across domains, yet traditional assessments focus on one domain only, failing to address cross-domain propagation. Purely safety or security risk assessment no longer meets the needs of ICS, requiring combined approaches. Machine learning is increasingly applied to anomaly detection and risk assessment in industrial settings, as evidenced by recent research on informed ML with graph attention networks and systematic reviews of ML for fault detection. These developments open the door for integrated, automated risk assessment solutions.

Sources: SIG-005CE36DA1FC Elsevier BV 2026-01-01; SIG-FA6096A63EB Elsevier BV 2026-01-01; SIG-03DDF2041F65 openalex.org 2026-02-03; SIG-EB5664191E1A openalex.org 2026-02-01

## Who buys, and why now

The buyer is typically the plant manager or the head of operational technology (OT) security, who feels the pain of manual compliance reporting and the fear of cross-domain incidents that could cause downtime or safety breaches. The trigger is often an upcoming regulatory audit or a recent near-miss incident that exposed the limitations of separate safety and security assessments. They need to demonstrate to regulators that risks are managed holistically, and they want to automate the reporting burden.

## Why it is hot now — every claim bound to a dated source

- A quantitative risk assessment method for industrial control system safety and security is needed to address cross-domain risk propagation. [SIG-005CE36DA1FC]

- Purely safety or security risk assessment no longer meets the needs of industrial control systems, requiring combined risk assessment. [SIG-FEA6096A63EB]

## What Orange would deliver, and why it can win

### What Orange would deliver

Orange would assemble a team of AI/Data/Cloud experts, Cybersecurity experts, and IoT experts to design and deploy a machine learning-based risk assessment platform. The engagement would start with a discovery phase using Orange Cyberdefense managed services to assess the current ICS environment, then deploy IoT sensors and leverage Live Objects for data ingestion. Machine learning models would be trained on historical data to detect anomalies and predict cross-domain risks, with outputs feeding automated compliance reports. Orange Group researchers would contribute to model development, and the solution would be hosted on a certified infrastructure (ISO 27001, TISAX) to meet sovereignty requirements.

### Why Orange can win

Orange can win because it combines deep cybersecurity expertise (Orange Cyberdefense) with IoT capabilities (Live Objects) and AI/Data/Cloud experts, all within a sovereign and certified framework (ISO 27001, TISAX). The ability to integrate safety and security domains, backed by Orange Group researchers, is a differentiator. However, the assets are not unique; hyperscalers and SI competitors also have strong ML offerings, so Orange must emphasize its end-to-end integration and trust credentials.

### Named assets behind this — each individually inspectable (LK-08)

| Asset                                | Type            | Link | Why it is linked                                           | Curator     |
|--------------------------------------|-----------------|------|------------------------------------------------------------|-------------|
| Live Objects                         | offer           | L1   | offer provides the technology in a matching domain         | unconfirmed |
| Orange Drone Guardian                | offer           | L1   | offer provides the technology in a matching domain         | unconfirmed |
| Orange Cyberdefense managed services | offer           | L1   | offer addresses the use case but not this technology       | unconfirmed |
| AWS                                  | partner         | L2   | partner provides the required technology                   | unconfirmed |
| Google Cloud                         | partner         | L2   | partner provides the required technology                   | unconfirmed |
| NICE                                 | partner         | L2   | partner provides the required technology                   | unconfirmed |
| NVIDIA                               | partner         | L2   | partner provides the required technology                   | unconfirmed |
| ISO 27001                            | certification   | SUP  | certification Orange holds is required by this vertical    | unconfirmed |
| TISAX                                | certification   | SUP  | certification Orange holds is required by this vertical    | unconfirmed |
| AI / Data / Cloud experts            | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Cybersecurity experts                | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| IoT experts                          | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Defense and security experts         | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Orange Group researchers             | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Colgate-Palmolive                    | reference       | SUP  | published reference in this vertical, different use case   | unconfirmed |
| Mondelez International               | reference       | SUP  | published reference in this vertical, different use case   | unconfirmed |

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

## Competitive landscape

**MEDIUM** competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

| Competitor | Category | Position here | Basis | Evidence |
|------------|----------|---------------|-------|----------|
|------------|----------|---------------|-------|----------|

|               |                              |                                                                                   |            |   |
|---------------|------------------------------|-----------------------------------------------------------------------------------|------------|---|
| AWS           | Hyperscaler                  | sells Machine learning; competes in OX: Smart Industries — also an Orange partner | structural | — |
| Google Cloud  | Hyperscaler                  | sells Machine learning — also an Orange partner                                   | structural | — |
| Atos / Eviden | Global systems integrator    | sells Machine learning; competes in Cybersecurity                                 | structural | — |
| NiCE          | Customer-experience platform | sells Machine learning — also an Orange partner                                   | structural | — |
| Databricks    | Data and AI platform         | sells Machine learning                                                            | structural | — |
| Dataiku       | Data and AI platform         | sells Machine learning                                                            | structural | — |
| Palantir      | Data and AI platform         | sells Machine learning; competes in OX: Smart Industries                          | structural | — |
| Accenture     | Global systems integrator    | active in Manufacturing; competes in OX: Smart Industries                         | structural | — |

**evidenced** means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

## Qualifying questions for the first meeting

- 1 How do you currently conduct safety and security risk assessments for your ICS, and are they integrated or separate?
- 2 What regulatory frameworks are you preparing for that require cross-domain risk reporting?
- 3 Have you experienced any incidents where a cyber event impacted physical safety or vice versa?
- 4 What is your current data collection and monitoring infrastructure for your industrial control systems?
- 5 Who in your organization owns the compliance reporting process, and how much time does it consume?
- 6 Are you open to a cloud-based solution, or do you have sovereignty requirements that mandate on-premise deployment?

## Objections you will meet

| They will say                                                                                | You can answer                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| We already have separate safety and security assessments; why do we need an integrated one?  | That's true, and they serve their purpose. But research shows that risks can propagate across domains, and regulators are increasingly expecting a holistic view. An integrated assessment can uncover gaps that separate assessments miss, and it can automate the reporting burden. |
| Machine learning sounds like a black box; how can we trust it for safety-critical decisions? | We understand the concern. That's why we combine ML with expert knowledge and human oversight. Our models are trained on your data and validated by our cybersecurity and OT experts, and we can provide explainability features to satisfy auditors.                                 |
| We're not sure about sharing our operational data with an external provider.                 | That's a valid concern. Orange offers sovereign and certified infrastructure (ISO 27001, TISAX) and can deploy on-premise if needed. We also have experience with large industrial clients who trust us with sensitive data.                                                          |

## Risks and unknowns

The main risk is that regulatory requirements may not yet mandate integrated safety and security assessments, so customers may not see an urgent need. Additionally, the effectiveness of ML models in this domain depends on data quality and availability, which may be limited in older ICS environments. There is also uncertainty about the willingness of manufacturers to share sensitive operational data with a third party, even a trusted one. Finally, the solution's complexity could lead to long sales cycles.

## Next action, by role (FR-17)

| Role                | Do this next                                                                                                                                                                                                                                                                                                                             |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Presales / Proposal | Prepare a bid brief that positions Orange Cyberdefense managed services and Live Objects as the foundation for a machine learning-driven risk assessment solution, referencing Saint-Gobain Glass as a proof point for manufacturing expertise, and outline how ISO 27001 and TISAX certifications support compliance.                   |
| Sales               | In a conversation with a manufacturing sector prospect's head of operations, say: 'We see that purely safety or security risk assessments no longer meet the needs of industrial control systems, and we're exploring how machine learning can automate compliance reporting for emerging regulations—would that be a priority for you?' |

|                        |                                                                                                                                                                                                                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Strategist / Innovator | Commission a deep-dive brief on the feasibility of a combined safety and security risk assessment framework for industrial control systems, leveraging Orange Group researchers and Cybersecurity experts to map cross-domain risk propagation and emerging compliance requirements. |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

| Signal           | Date       | Publisher                    | T | Title and link                                                                                     |
|------------------|------------|------------------------------|---|----------------------------------------------------------------------------------------------------|
| SIG-36DA075F2A6A | 2026-03-02 | openalex.org                 | 1 | IoT-based Industrial Safety and Monitoring System with Real-Time Alert Mechanism                   |
| SIG-03DDF2041F65 | 2026-02-03 | openalex.org                 | 1 | Anomaly detection for industrial time series in process industry using informed machine learnin... |
| SIG-EB5664191E1A | 2026-02-01 | openalex.org                 | 1 | A Systematic Review of Anomaly and Fault Detection Using Machine Learning for Industrial Machin... |
| SIG-55299B90A514 | 2026-01-14 | Industrial Automation and... | 1 | Industrial Automation and Control System Security Training                                         |
| SIG-74751753706B | 2026-01-14 | Industrial Automation and... | 1 | Emerging Approaches to Industrial Automation and Control System Security                           |
| SIG-920373EC23E1 | 2026-01-14 | Industrial Automation and... | 1 | Risk Management for Industrial Automation and Control Systems                                      |
| SIG-A5B27741EBA6 | 2026-01-14 | Wiley                        | 1 | Industrial Automation and Control System Security Principles                                       |
| SIG-005CE36DA1FC | 2026-01-01 | Elsevier BV                  | 1 | A quantitative risk assessment method for industrial control system safety and security            |
| SIG-DBD887DDFE0D | 2026-01-01 | IEEE Internet of Things J... | 1 | Security for Safety Collaboration Strategy in Industrial Control System                            |
| SIG-FEA6096A63EB | 2026-01-01 | Elsevier BV                  | 1 | A Survey on Key Issues of Risk Assessment methods for Industrial Control System Security and Sa... |
| SIG-A33E4A343E60 | 2025-12-17 | openalex.org                 | 1 | Immersive technologies for evaluating industrial safety training in high-risk environments: a r... |
| SIG-6ADB457BBC36 | 2025-11-19 | openalex.org                 | 1 | A Machine Learning Approach to Predicting Air Quality and Health Risks for Construction Worker ... |
| SIG-7ADD7236CDF2 | 2025-09-01 | openalex.org                 | 1 | Preliminary Design Guidelines for Evaluating Immersive Industrial Safety Training                  |
| SIG-B18581CE4292 | 2025-04-25 | 2025 IEEE 3rd Internation... | 1 | Research on the Intelligent System of Industrial Control Network Security Protection Based on A... |

## How this brief was made

**Computed, not written.** Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

**Written, then checked.** The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive\_landscape (no claim survived evidence binding); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

| What                        | Version                     | When                      |
|-----------------------------|-----------------------------|---------------------------|
| Scores — weight set         | w-2026-08-a                 | 2026-08-18T19:08:26+00:00 |
| Market sizing — assumptions | size-2026-08-a              | 2026-08-18T21:06:16+00:00 |
| Competitor register         | competitors-2026-08-a       | 2026-08-18T21:06:19+00:00 |
| Narrative — prompt / model  | describe-v1 / deepseek-chat | 2026-08-19T04:56:59+00:00 |
| Pipeline                    | 0.1.0                       | 2026-08-18                |

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.